

Applied Generative AI

Prompt Engineering: Basics

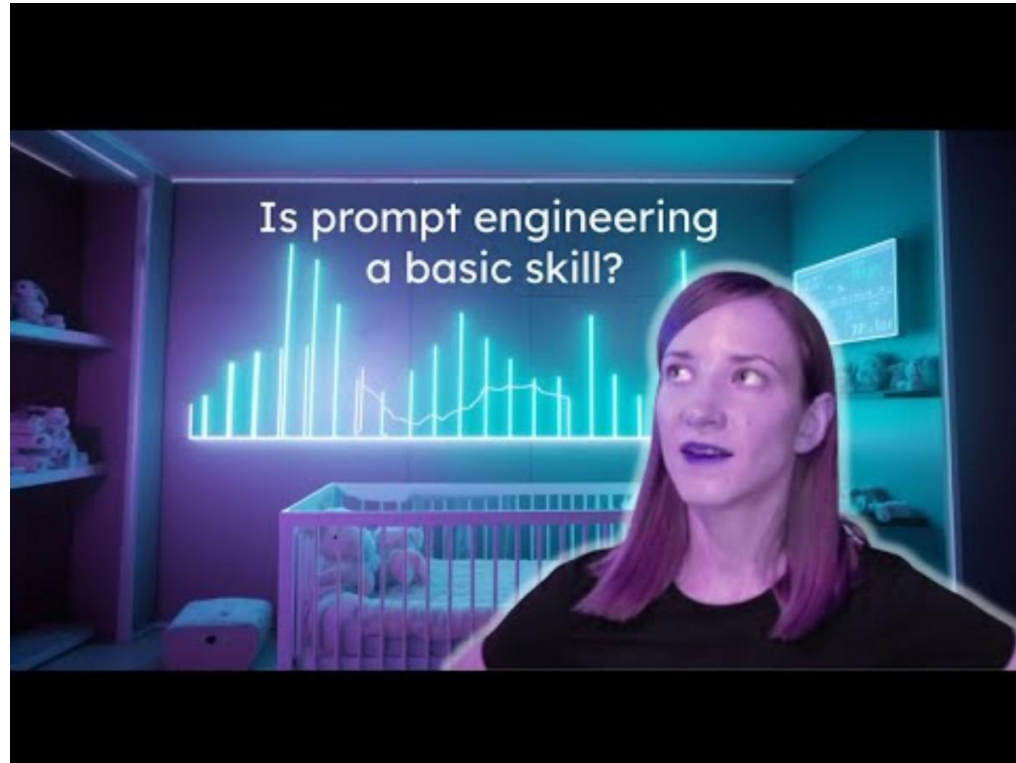
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Learning Objectives

- Understand what prompt engineering is
- Learn the basics of prompt engineering and common design patterns
- Use LangChain for Prompt Engineering

What is Prompt Engineering?



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Kozyrkov, Cassie, *Is prompt engineering a basic skill? Is it even... engineering?*

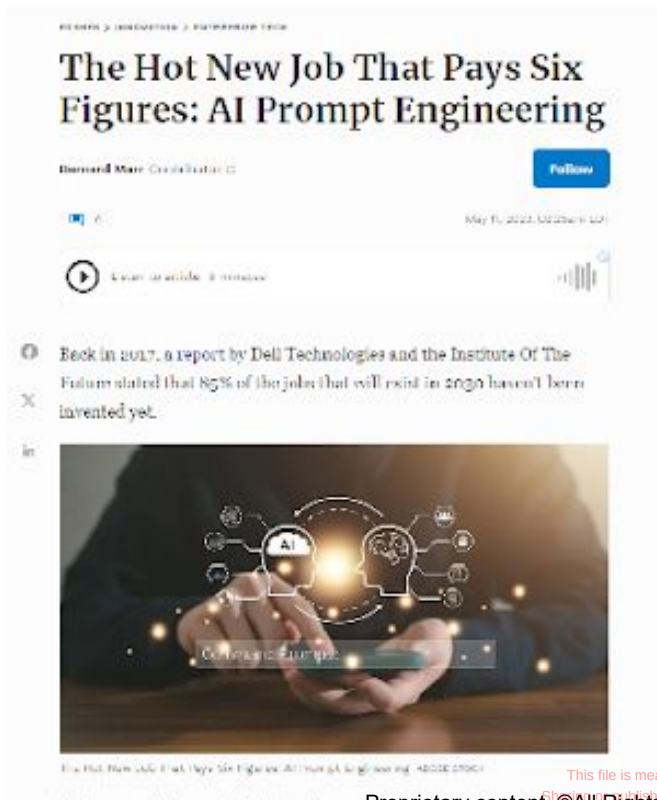
<https://www.youtube.com/watch?v=L2jnRK2GYwg&t=296s>

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What is Prompt Engineering?

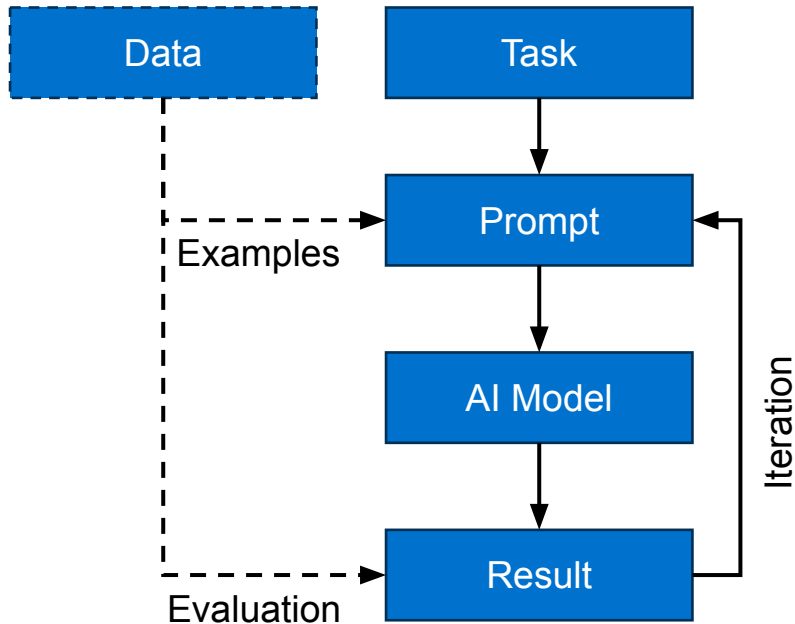
- *Prompt Engineering* is the art and science of designing and structuring *prompts* (questions or tasks) fed to language models.
- It is a collection of strategies and methods to describe a task in text for an AI model.
- Improve the performance of the model on tasks.

Why Do We Engage in Prompt Engineering?



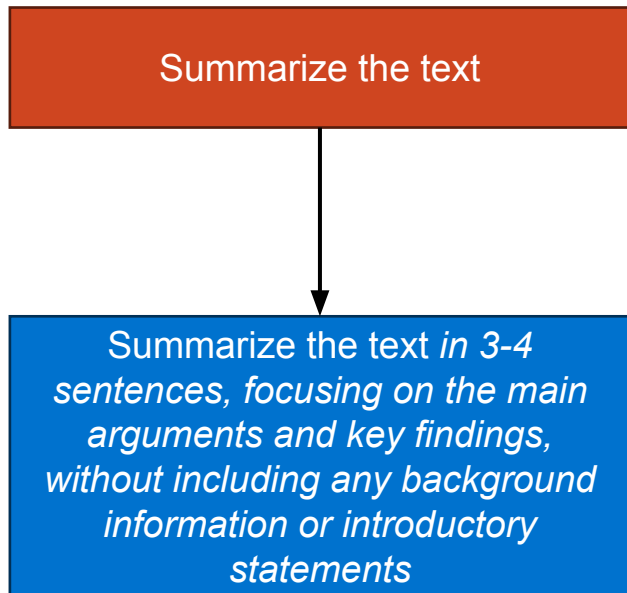
- To adapt the model to a wide range of tasks and applications, offering flexibility.
- To guide the model's responses, leading to more accurate and relevant results.
- To get the desired output efficiently, saving computational resources.

The Basic Elements of an Effective Prompt



- Effective prompts normally have a couple of basic elements:
 - Clear instruction
 - Context
 - Output indicator
- Prompting frequently also benefits from additional elements:
 - Examples
 - Embodiment
 - Task breakdown

Clear Instructions in a Prompt



- Clearly define the task you want the model to do.
- Provide necessary details to do the task.
- Avoid vague language.
- For difficult to define tasks, providing examples (i.e., few-shot prompting) can help.

Adopting a Persona in a Prompt

Analyze and describe the tone of this passage in a few sentences



You are literary expert who focuses on early medieval literature.
Analyze and describe the tone of this passage in a few sentences

- Have the AI model play a role.
- The role should be related to the task you want the model to do.
- Providing descriptive elements to the role can also improve performance (e.g., “experienced”, “from [location]”, etc.).

Adding Context to a Prompt

"Why is cybersecurity important in healthcare?"

"Why is cybersecurity important in healthcare?
Consider this context: Healthcare systems store sensitive patient information,"

Using this context, explain why cybersecurity is especially crucial in the healthcare industry.

- Provide additional details to the prompt.
 - Forms the basis of Retrieval Augmented Generation.
- Greatly helps with problems like Hallucination.
- Can be used to augment AI models with knowledge outside of their training.

Adding Text Delimiters to a Prompt

Answer the question based on this passage

Based on the *[Passage]* provided, answer the *[Question]* accurately and concisely:
[Passage]: " ..."
[Question]: "What are some of the fields impacted by recent AI developments?"
[Answer]:

- Provide textual cues for elements of the prompt.
 - Task
 - Examples
 - Output indicator
- Cue words like "example:" or "output:" can mark particular actions for the model.
- Delimiters like "..." or [...] can demarcate special sections of text.

Breaking Down a Task in a Prompt

Analyze the following passage for tone, main ideas, and any notable rhetorical devices.

Analyze the following passage by completing each step below:

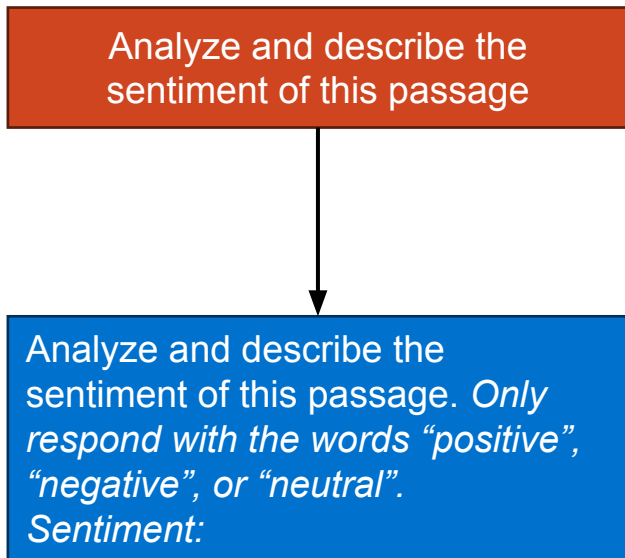
1. *Identify and describe the tone of the passage in one sentence.*
2. *List the two main ideas conveyed in the passage.*
3. *Highlight any rhetorical devices (e.g., metaphors, alliteration) used, and briefly explain their effect.*

- Specify the steps required to do a task.
- Break the task into different sub-tasks which can be done over successive prompts.
- For tasks with lots of different instruction sets, classify a type of query and then use the appropriate instruction set.

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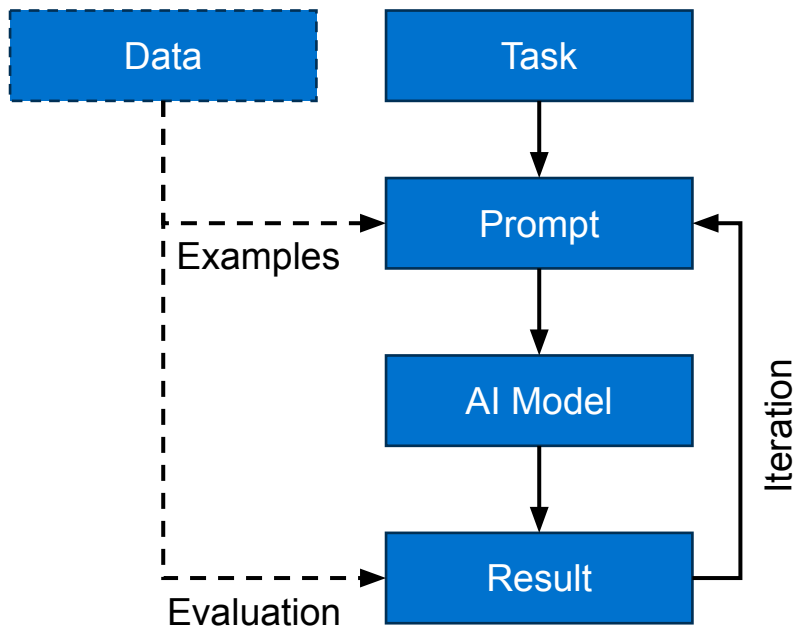
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Specifying Output in a Prompt



- Be descriptive about the output desired.
 - Word or sentence length
 - Special words to use (i.e., in a classification task)
- Provide an output indicator or word to cue the model (e.g., ":").

Bringing it All Together



- Prompt engineering is often an iterative process.
 - It helps to have some kind of validation data of what you are looking for.
 - Test changes systematically.
- Good prompts frequently use all or many of the aspects previously described.
- Ultimately, you want to structure your inputs to AI models to optimize your outputs from them.



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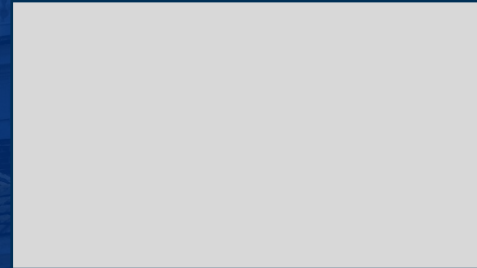
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Prompt Engineering: Common Design Patterns



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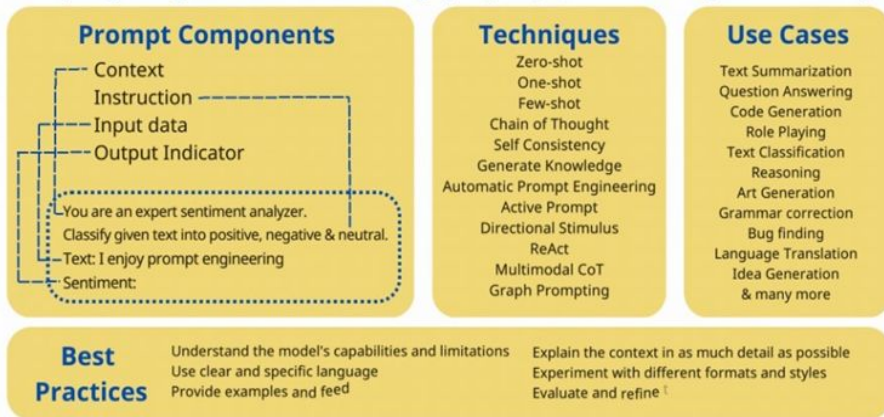
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Common Prompting Design Patterns

Prompt Engineering

(Effective communication & collaboration with AI)

Prompt Engineering is art and science of crafting inputs(prompts) to AI models to get the desired output



Vaj, Tiya, *Prompt Engineer* (2024)

- There are a number of prompt formats, or design patterns, for various types of tasks.
- Some of the most successful feature:
 - Using examples and/or context data.
 - Having the AI model engage in some form of deliberate reasoning.

Few Shot Prompting

The diagram shows a prompt structure for a translation task. It consists of five lines of text, each with a number on the left. The first line is the task description. The next three lines are examples of input-output pairs. The fifth line is the prompt to be completed. Arrows on the right point from labels to the corresponding lines: 'task description' points to line 1, 'examples' points to lines 2, 3, and 4, and 'prompt' points to line 5.

```
1  Translate English to French:
2  sea otter => loutre de mer
3  peppermint => menthe poivrée
4  plush girafe => girafe peluche
5  cheese => .....
```

Brown et al., *Language Models are Few-Shot Learners* (2020)

- Provide the model with several examples of the task.
- Example: For a translation task, provide multiple pairs of sentences in two languages.
- Cautions on use:
 - The order and choice of examples can greatly influence the outcome.

Chain-of-Thought Prompting

Standard Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

Model Output

A: The answer is 27. ❌

Chain of Thought Prompting

Input

Q: Roger has 5 tennis balls. He buys 2 more cans of tennis balls. Each can has 3 tennis balls. How many tennis balls does he have now?

A: Roger started with 5 balls. 2 cans of 3 tennis balls each is 6 tennis balls. $5 + 6 = 11$. The answer is 11.

Q: The cafeteria had 23 apples. If they used 20 to make lunch and bought 6 more, how many apples do they have?

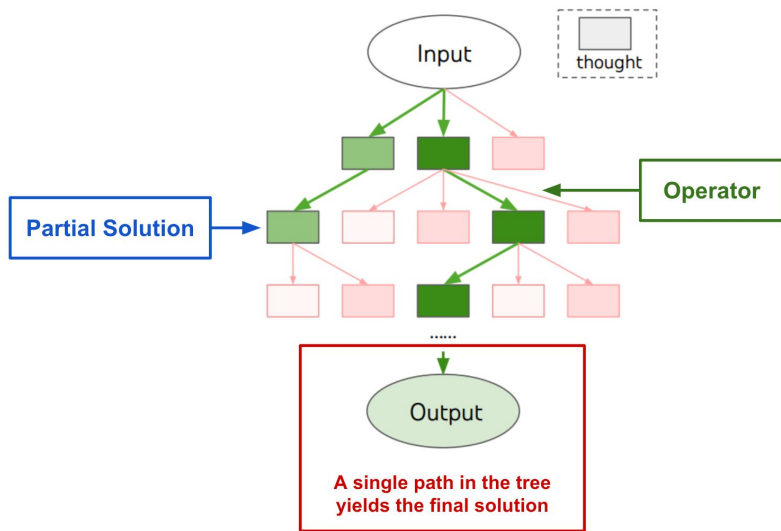
Model Output

A: The cafeteria had 23 apples originally. They used 20 to make lunch. So they had $23 - 20 = 3$. They bought 6 more apples, so they have $3 + 6 = 9$. The answer is 9. ✅

- Break down complex tasks into a series of simpler tasks, guiding the model through a chain of reasoning.
- Example: For an arithmetic task, start with simple operations and build up to the more complex operation.
- Cautions on use:
 - Increased computation cost
 - Design of intermediate steps

Wei et al., *Chain-of-Thought Prompting Elicits Reasoning in Large Language Models* (2022)

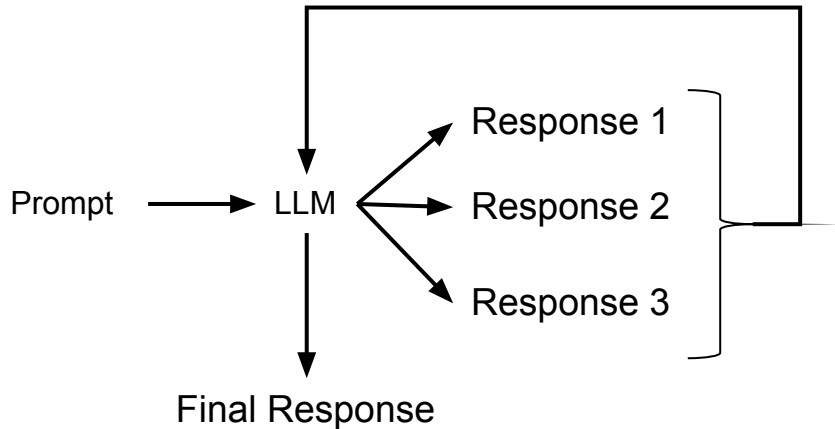
Tree-of-Thought Prompting



- Guide the model to explore multiple reasoning paths by branching into different lines of thought.
- Example: For solving a logic puzzle, prompt the model to generate multiple hypotheses and evaluate each one before converging on a solution.
- Cautions on use:
 - Increased computation cost (more than CoT)
 - Design of merge and prune steps

Yao et al., *Chain-of-Thought Prompting Elicits Reasoning in Large Language Tree of Thoughts: Deliberate Problem Solving with Large Language Models* (2023)

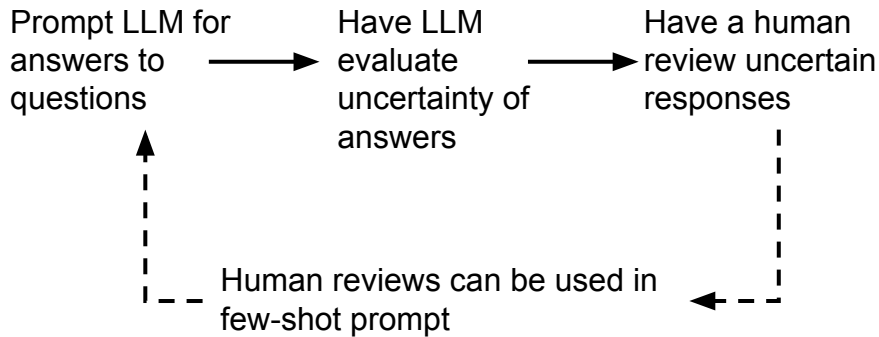
Self-Consistency Prompting



- Reminded the AI model of previous parts of the conversation or ask it to ensure its responses are consistent with prior statements.
- Example: For a question-answering task, produce several possible answers and then seek consistency across the answers.
- Cautions on use:
 - Increased computation cost
 - Risk of Majority Bias

Wang et al., *Self-consistency improves chain-of-thought reasoning in language models* (2023)

Active Prompting



- AI model proactively seeks additional information or clarification from the user to improve its understanding of the task or prompt.
- Example: Having the model ask questions of the user about the how to classify something before doing the task
- Cautions on use:
 - Relies on human expertise

Diao et al., *Active Prompting with Chain-of-Thought for Large Language Models* (2023)

Action-Observation Prompting

(1d) ReAct (Reason + Act)

Thought 1: I need to search Apple Remote and find the program it was originally designed to interact with.

Act 1: Search[Apple Remote]

Obs 1: The Apple Remote is a remote control introduced in October 2005 by Apple ... originally designed to control the Front Row media center program ...

Thought 2: Apple Remote was originally designed to control the Front Row media center program. I need to search Front Row next and find what other device can control it.

Act 2: Search[Front Row]

Obs 2: Could not find [Front Row]. Similar: ['Front Row Seat to Earth', 'Front Row Motorsports', 'Front Row (software)', ...]

Thought 3: Front Row is not found. I need to search Front Row (software) .

Act 3: Search[Front Row (software)]

Obs 3: Front Row is a discontinued media center software ...

Thought 4: Front Row (software) is controlled by an Apple Remote or the keyboard function keys. So the answer is keyboard function keys.

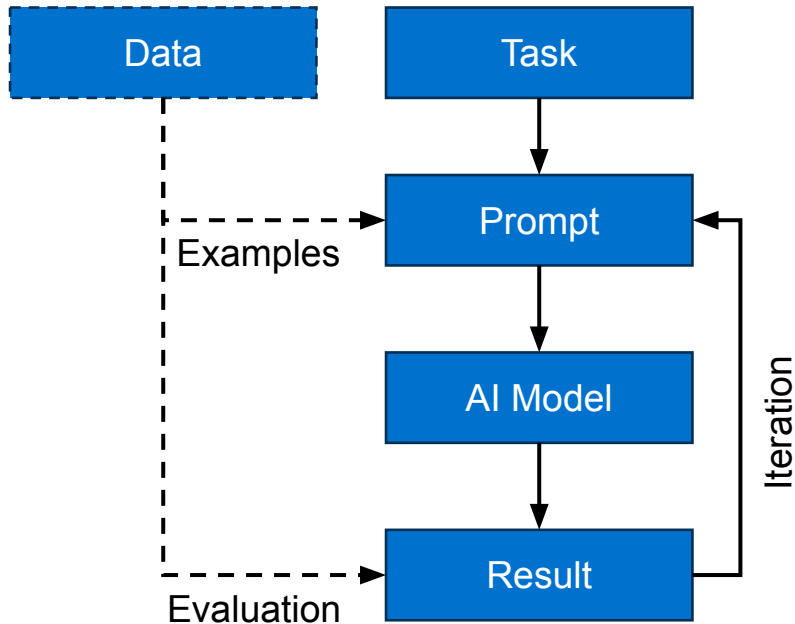
Act 4: Finish[keyboard function keys]



- Alternates between performing actions (e.g., reasoning steps) and observing outcomes, using feedback to refine responses.
- Example: For a question-answering task, the model forms a hypothesis, retrieves relevant data, adjusts based on findings, and refines the answer.
- Cautions on use:
 - Complexity in prompt design

Yao et al., ReAct: Synergizing Reasoning and Acting in Language Models (2022)

Evaluating the Quality of a Prompt



- Evaluation against performance indicators
 - Consistency
 - Completeness and specificity of the output
 - Comparison to benchmarks
- Test with variations and red-teaming
 - Ambiguous or harmful input
 - Especially for user facing applications
- Iterative refinement



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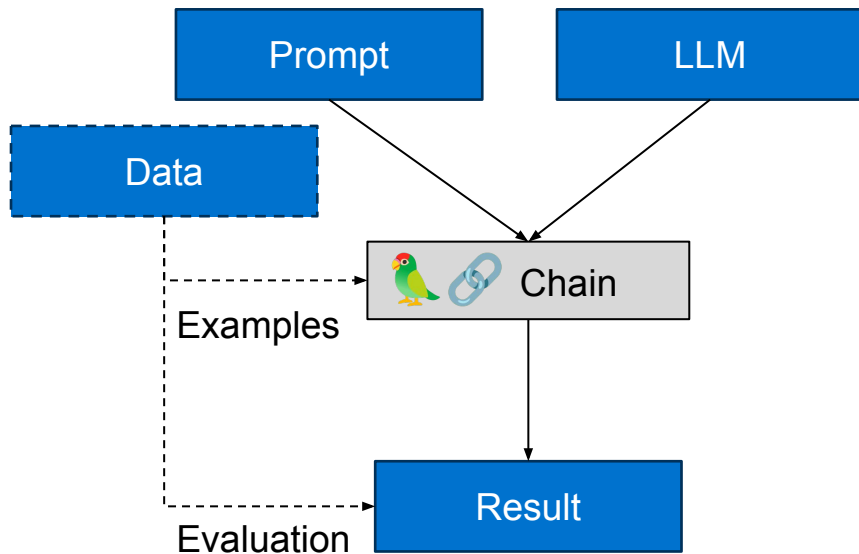
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Prompt Engineering: Working with LangChain  

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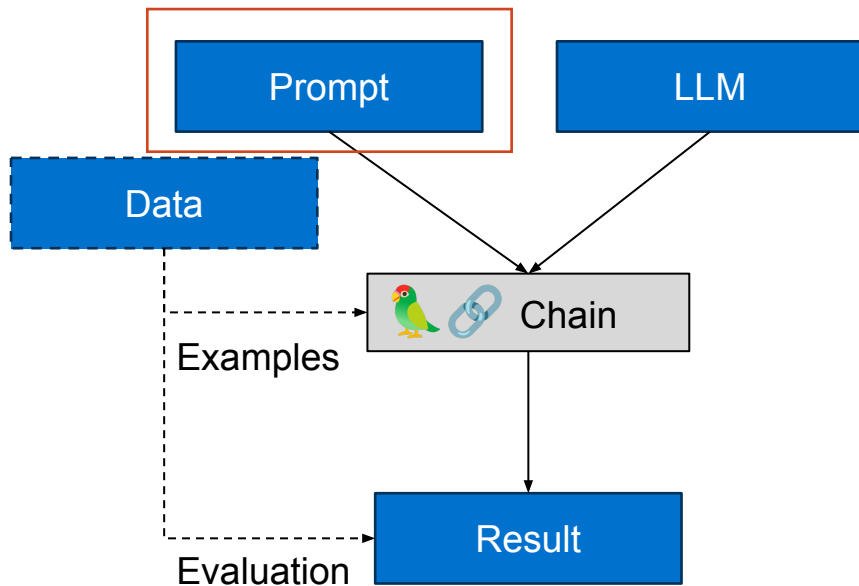
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LangChain Core Concepts



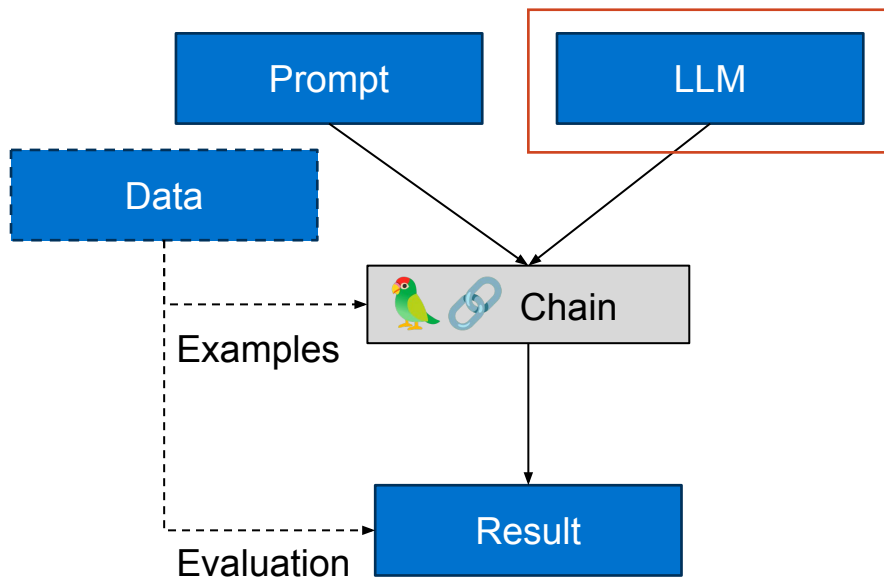
- LangChain uses the concept of “Chains” to link together things like models and prompts, and control their execution.
- For prompt engineering, the components of our chains will center around:
 - Prompt templates
 - LLMs
 - Chains

Prompt Templates



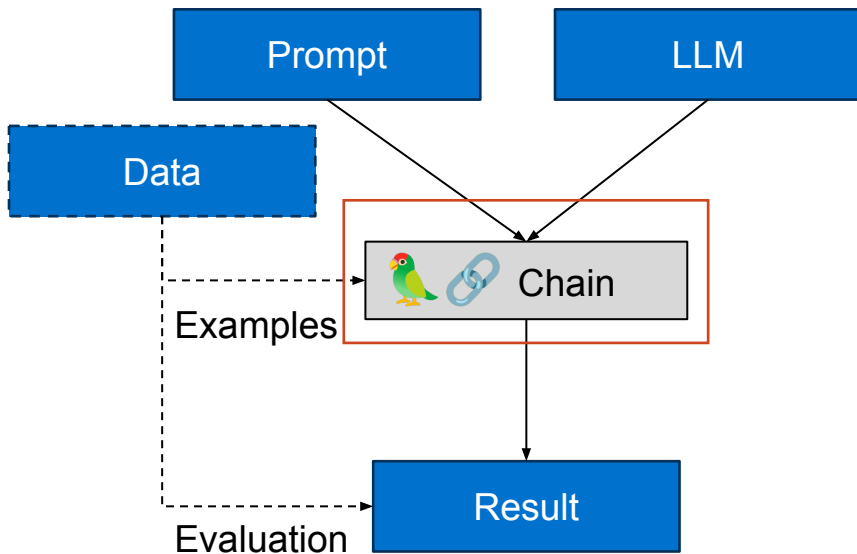
- Prompts are stored in Prompt templates, which consist of:
 - Text for the prompt with '{ }' for inserting variables
 - A Prompt Template object that controls inputs and outputs
- There are a number of Prompt Template classes, but the most important are the basic `BasePromptTemplate` and `FewShotPromptTemplate`

LLMs



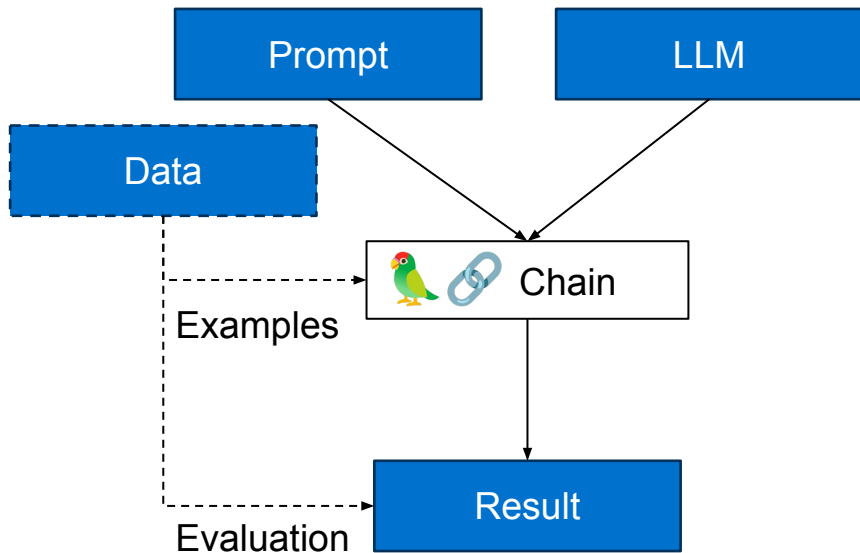
- LangChain uses an abstraction to wrap models for inclusion into Chains.
- Typically each major source of LLMs (i.e., OpenAI, Cohere, Huggingface) has a specific class for wrapping those models (or API calls to those models)
 - Model classes are also typically broken between “chat” and “llm” classes depending on the particular model

Chains



- Chains are the reusable components that link together things like LLMs and prompts
- Chains are composable with other objects or “runnables”, including other chains
- Chains typically link together runnables through the “|” operator

A Final Note about LangChain



- In this section, we focused in on the elements of LangChain that most directly apply to prompt engineering
- LangChain has many other elements for things like Agents, memory, and API calls, which can also be incorporated in chains
- I invite you to explore the documentation to find other chains or components of chains!



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